LECTURE NOTES: LIMITS AND CONTINUITY

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Abstract. A brief introductory exposition on limits, continuity, and related theorems.

1. Limit of a Function

1.1. Examples of Limits.

We begin with an intuitive definition of limits of real-valued functions.

Definition 1.1. For a real-valued function $f : D \to \mathbb{R}$ defined on some subset $D \subseteq \mathbb{R}$, we say that *the limit of $f(x)$ as x approaches a is L* if the value of $f(x)$ gets arbitrarily close to L as x gets closer and closer to a, without necessarily reaching a.

That is, the existence of a limit depends only on the behavior of the function near the point a, regardless of whether $f(a)$ is defined. In particular, the limit $\lim_{x\to a} f(x)$ exists if and only if any sequence of numbers $\{x_n\}$ approaching a within some neighorhood of a produces a sequence of numbers $\{f(x_n)\}$ approaching the same value L. Since we can approach a from both sides (either right or left) on the real number line, for the overall limit to exist, the *right-hand limit* $\lim_{x\to a^+} f(x)$ and *left-hand limit* $\lim_{x\to a^-} f(x)$ must both exist and equal the same value L.

Here we look at some "nontrivial" examples (or counterexamples) of limits of functions.

Example 1.2. Let $f(x) = \sin\frac{1}{x}$. If we ever try to estimate the limit $\lim_{x\to 0} f(x)$ by either graphing or calculating $f(x)$ for very small values of x, we quickly run into an issue; the functions seems to "fluctuate" across many different values as we take $x = 0.01, 0.00001, 0.0000001$, and so on.

In fact, the function indeed oscillates *infinitely often* on any open interval containing the point 0. In particular, if we let

$$x_1 = \frac{1}{\frac{1}{2}\pi + 2n\pi}, \qquad x_2 = \frac{1}{\frac{3}{2}\pi + 2m\pi},$$

then any open interval around 0 contains the points x_1 and x_2 for sufficiently large n and m, while $f(x_1) = 1$ and $f(x_2) = -1$, meaning that the function oscillates between these two values (which are indeed upper and lower bounds of f).

Example 1.3. Let $f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$

Since there is always an irrational number arbitrarily close to any real number, the function oscillates between 0 and 1 infinitely often on any given interval on $\mathbb{R}$ and the limit $\lim_{x \to a} f(x)$ does not exist for any $a \in \mathbb{R}$. This function is also known as the *Dirichlet function.*

Example 1.4. Let $f(x) = \begin{cases} \frac{1}{q} & \text{if } x = \frac{p}{q} \text{ in lowest terms}, \\ 0 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$

Here f is commonly known as the *Thomae function* or the *popcorn function.* Take some $a \in (0, 1)$. Then for any sequence of real numbers $\{x_n\}$ approaching a, we can split the sequence in two parts: (1) a sequence of irrational numbers $\{z'_n\}$ approaching a, and (2) a sequence of rational numbers $\{z_n\}$ approaching a. The first sequence $\{z'_n\}$ of irrational numbers produces a sequence of function values $\{f(z'_n)\}$ that approaches 0.

The second sequence $\{z_n\}$ requires a little more thinking. If we write $z_n = \frac{p_n}{q_n}$ in lowest terms, then the function values $f(z'_n)$ form a sequence of form $\{\frac{1}{q_n}\}$. Now the question becomes: can we find a sequence of rational numbers $\{z_n\}$ approaching a such that the denominators q_n are bounded above by some constant M? If not, then the function values $f(z_n) = \frac{1}{q_n}$ must approach 0. The answer is no; if we take M to be any positive integer, then there are only finitely many rational numbers in $(0, 1)$ with denominators at most M, so we cannot find a sequence of rational numbers approaching a with bounded denominators. Thus the function values $f(z_n)$ must approach 0 as well, and we conclude that $\lim_{x \to a} f(x) = 0$ for any $a \in (0, 1)$.

Example 1.5. Now let $f(x) = \frac{\sin \theta}{\theta}$ and consider the limit of $f(\theta)$ as θ approaches 0. Geometrically, $\sin \theta$ is the length of the vertical segment of the unit vector at angle θ from the positive x-axis. We see that the triangle formed by the origin, the point on the unit circle at angle θ, and the unit vector $(1, 0)$ has area $\frac{1}{2} \sin \theta$, while the sector of the unit circle at angle θ has area $\frac{1}{2}\theta$. Since the triangle is contained in the sector, we have

$$\frac{\sin \theta}{2} \leq \frac{\theta}{2}, \quad \text{for } 0 < \theta < \frac{\pi}{2}.$$

On the other hand, the triangle formed by the origin, the unit vector $(1, 0)$, and the point $(1, \tan \theta)$ has area $\frac{1}{2}$. Since this larger right triangle contains the sector, we have the inequality

$$\frac{\theta}{2} \leq \frac{\tan \theta}{2}.$$

Combining the two inequalities, we have

$$\frac{\sin \theta}{2} \leq \frac{\theta}{2} \leq \frac{\tan \theta}{2}.$$

Multiplying through by $\frac{2}{\sin\theta}$, we have

$$1 \le \frac{\theta}{\sin\theta} \le \frac{1}{\cos\theta}.$$

for $\theta \in (0, \pi/2)$. Thus $\lim_{x\to 0^+} \frac{\sin\theta}{\theta} = 1$.

To show that $\lim_{x\to 0^-} \frac{\sin\theta}{\theta} = 1$, notice that $\sin\theta$ is an odd function. Thus after changing the sign of θ we obtain

$$\begin{aligned}\lim_{x\to 0^-} \frac{\sin\theta}{\theta} &= \lim_{x\to 0^+} \frac{\sin(-\theta)}{-\theta} \\ &= \lim_{x\to 0^+} \frac{-\sin\theta}{-\theta} \\ &= \lim_{x\to 0^+} \frac{\sin\theta}{\theta} \\ &= 1.\end{aligned}$$

Thus we have $\lim_{x\to 0} \frac{\sin\theta}{\theta} = 1$.

1.2. **The Squeeze Theorem.** So far, we have seen examples of limits of non-polynomial or non-rational functions that require a more sophisticated definition of a limit. We continue to examine limits of functions that cannot be computed directly through standard algebraic manipulation.

Theorem 1.6 (Squeeze Theorem)**.** *Let f, g, h be real-valued functions defined on some open interval containing a, except possibly at a itself. If $f(x) \le g(x) \le h(x)$ for all x in the interval, and if $\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L$, then $\lim_{x\to a} g(x) = L$ as well.*

Essentially, we are trapping the tricky function g between two functions f and h whose limits are well-defined and have the same value. In that case, the limit of the trapped function g is forced to approach the same value as well.

Example 1.7. Consider the limit $\lim_{x\to 0} x \sin\frac{1}{x}$. Notice that we cannot algebraically compute this limit since

$$\lim_{x\to 0} \sin\frac{1}{x}$$

does not exist (example 1.2). We can apply the squeeze theorem by noting that $-1 \le \sin\frac{1}{x} \le 1$ for all x, so we have

$$-x \le x\sin\frac{1}{x} \le x.$$

Since $\lim_{x\to 0} -x = \lim_{x\to 0} x = 0$, we conclude that $\lim_{x\to 0} x\sin\frac{1}{x} = 0$ as well even though the limit of $\sin\frac{1}{x}$ does not exist.

2. Continuity

2.1. **Examples of Continuous Functions.** We now begin the discussion of *continuous* functions. Throughout the course, the analysis of functions that are continuous on their domains will be the main theme.

Definition 2.1. A real-valued function f is called *continuous* at point a if $\lim_{x\to a} f(x) = f(a)$.

From this pointwise definition of continuity the following definitions of continuity on an interval follow immediately.

Remark 2.2. f is continuous on an open interval (a, b) if f is continuous at every point in (a, b). A special case of this is when f is continuous on $\mathbb{R}$, in which case f is continuous on $(-\infty, \infty)$.

Remark 2.3. On a closed interval $[a, b]$, we have to be slightly more careful since the endpoints a and b do not have neighborhoods that are entirely contained in $[a, b]$. Thus we say that f is continuous on $[a, b]$ if f is continuous at every point in (a, b), and if $\lim_{x\to a^+} f(x) = f(a)$ and $\lim_{x\to b^-} f(x) = f(b)$.

Now we describe some examples of continuous and discontinuous functions to help illustrate the definition of continuity.

Example 2.4. As we saw in example 1.7, the function $f(x) = x \sin \frac{1}{x}$ is undefined at $x = 0$. In fact, f is continuous everywhere else. If we define a new function $g(x)$ by

$$g(x) = \begin{cases} x \sin \frac{1}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0, \end{cases}$$

then g is continuous at 0 since $\lim_{x\to 0} g(x) = \lim_{x\to 0} x \sin \frac{1}{x} = 0 = g(0)$.

In such a case where we can define a continuous function g by assigning a value to f at the point of discontinuity, we say that f has a *removable discontinuity* at that point. More formally, we say that f has a removable discontinuity at a if the limit $\lim_{x\to a} f(x)$ exists but does not equal $f(a)$ (or $f(a)$ is undefined).

Example 2.5. On the other hand, for $f(x) = \sin \frac{1}{x}$, there is no value a such that the new function

$$g(x) = \begin{cases} \sin \frac{1}{x} & \text{if } x \neq 0, \\ a & \text{if } x = 0, \end{cases}$$

is continuous since the limit $\lim_{x\to 0} \sin \frac{1}{x}$ does not exist (example 1.2).

We say that f has a *jump discontinuity* at a if the left-hand and right-hand limits $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ both exist but are not equal to each other.

2.2. **The Intermediate Value Theorem.**

Theorem 2.6. *If f is ontinuous on [a, b] and $f(a) < c < f(b)$, then there exists $x \in [a, b]$ such that $f(x) = c$.*

The intermediate value theorem states that a continuous function on a closed, bounded interval takes all values between the two values on the boundary. We give a few examples to illustrate the theorem.

Example 2.7. Let $f(x) = x^3$. Then f has a root in the closed interval $[-1, 1]$ since f is continuous on $[-1, 1]$ and $f(-1) = -1$ while $f(1) = 1$.

Example 2.8. Let $f : [a, b] \to [a, b]$ continuous. We will show that f has a *fixed point*, i.e., there exists $c \in [a, b]$ such that $f(c) = c$. Let $g(x) = f(x) - x$. We want to show that $g(c) = 0$ for some $a \leq c \leq b$. Since f maps the interval $[a, b]$ back to itself, $a \leq f(x) \leq b$ for any $x \in [a, b]$. Thus $0 \leq g(a)$ and $0 \geq g(b)$. If either $g(a)$ or $g(b)$ is zero we are done. Otherwise, $g(a) > 0$ and $g(b) < 0$ so by the intermediate value theorem it immediately follows that f has a fixed point.

Example 2.9. Let

$$f(x) = a_d x^d + a_{d-1} x^{d-1} + \cdots + a_1 x + a_0$$

be an odd degree polynomial (the degree d is odd). We will show that f always has at least one root in $\mathbb{R}$. First notice that

$$\lim_{x \to \infty} f(x) = \infty, \quad \lim_{x \to -\infty} f(x) = -\infty.$$

Thus for any $y \in \mathbb{R}$, there exists R such that

$$x > R \implies f(x) > y, \quad x < R \implies f(x) < y.$$

By the intermediate value theorem it follows that there is some $x_0 \in [-R, R]$ such that $f(x_0) = y$ for any $y \in \mathbb{R}$.

Example 2.10. Any even degree polynomial attains either a minimum or maximum on $\mathbb{R}$. Proof is left as an exercise for the reader.